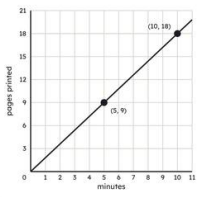
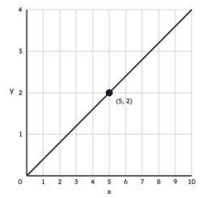
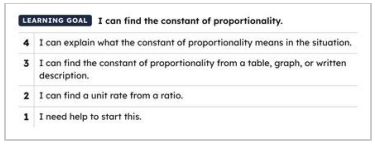
# C-AR.4.2 · Weekly check · 5 questions

MA.7.AR.4.2 · Constant of proportionality · Unit 11 Representing Proportional Relationships · Given Thursday, on the week's benchmark

**Form title** C-AR.4.2 Check · Constant of proportionality
**Description** Learning goal: I can find the constant of proportionality. 5 questions, 1 point each. Work each one out on paper, then choose or type your answer. Calculator allowed.
**Locked mode** OFF (calculator allowed; Locked mode blocks the Chromebook calculator)

**Short-answer validation** (already on the template's short-answer question; copy that question): Regular expression → Matches → `^-?(\d+(\.\d+)?|\.\d+|\d+/\d+|\d+ \d+/\d+)$`
Error text: Type only a number, like 12 or -3.5 or 2/7 or 4 1/3. No units, $, %, π or commas.

| # | Question image | File · type | Options to add | Answer key | Pts |
|---|---|---|---|---|---|
| 1 | An irrigation system releases 64.05 cups of water in 7 minutes at a constant rate. Let $x$ be the number of minutes and let $y$ be the number of cups of water released.<br>What is the constant of proportionality, $k$, in $y = kx$?<br>A. 64.05<br>B. 57.05<br>C. 9.15<br>D. 0.11 | **Q1.png**<br>Multiple choice<br>Same problem, now multiple choice (paper: multiple choice) | A B C D | **C** | 1 |
| 2 | The daily fee for docking a boat at a marina is proportional to the length of the boat. The table shows the fee for four different boat lengths.<br>Boat length (feet) \| Daily fee<br>20 \| $62.00<br>22 \| $68.20<br>24 \| $74.40<br>28 \| $86.80<br>What is the constant of proportionality, in dollars per foot?<br>Type a number only. No units, $ or %. | **Q2.png**<br>Short answer | none (typed answer) | **3.1 · 3.10**<br>Tick "Mark all other answers incorrect."<br>Response validation: Regular expression, Matches (see top). | 1 |
| 3 | The graph shows the proportional relationship between minutes and pages printed.<br>What is the constant of proportionality, in pages per minute?<br>  <br>Type a number only. No units, $ or %. | **Q3.png**<br>Short answer | none (typed answer) | **1.8 · 1.80 · 1 4/5 · 9/5**<br>Tick "Mark all other answers incorrect."<br>Response validation: Regular expression, Matches (see top). | 1 |
| 4 | The graph shows a proportional relationship between $x$ and $y$.<br>What is the constant of proportionality? Write your answer as a fraction in simplest form.<br>  <br>Type a whole number, a decimal like 0.25, a fraction like 2/7 or -3/4, or a mixed number like 4 1/3 with a space after the whole number. | **Q4.png**<br>Short answer | none (typed answer) | **2/5**<br>Tick "Mark all other answers incorrect."<br>Response validation: Regular expression, Matches (see top). | 1 |
| 5 | Each batch of paint is made with 2 cups of blue paint and 12 cups of white paint. The relationship between cups of blue paint and cups of white paint is proportional. Let $x$ be the number of cups of white paint and let $y$ be the number of cups of blue paint.<br>What is the constant of proportionality, and what does it mean in this situation?<br>A. 2; each batch uses 2 cups of blue paint.<br>B. 6; each cup of blue paint is used with 6 cups of white paint.<br>C. $\frac{1}{6}$; each cup of blue paint is used with $\frac{1}{6}$ cup of white paint.<br>D. $\frac{1}{6}$; each cup of white paint is used with $\frac{1}{6}$ cup of blue paint. | **Q5.png** Rigor<br>Multiple choice<br>Same problem, now multiple choice (paper: a written answer) | A B C D | **D** | 1 |

| | | | | | |
|---|---|---|---|---|---|
| **6** |  | **Level scale.png**<br>Multiple choice (the template's last question) | Level 1 Level 2 Level 3 Level 4 | no key | 0 |

Answers with worked solutions are in the Answers folder next to the question images (same file names).